

ALI DBOUK

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Quantitative engineer with experience owning and applying simulation-driven models and analytics to support high-stakes decision-making in high-pressure environments, including Mercedes F1. Proven track record of translating large-scale data into actionable recommendations, optimising risk-reward trade-offs, and improving decision accuracy and speed through automation, predictive modelling, and cross-system integration. Strong in Python, machine learning, and systems design, with a bias toward execution and measurable impact.

Tech Stack: Python, SQL, FastAPI, Pandas/NumPy, TensorFlow, Langgraph, OpenCV, AWS (EC2, S3, Lambda), Docker, Git, MATLAB, Kubernetes

Employment

Mercedes-AMG Petronas Formula One Team

Brackley, England

Performance and Simulation Engineer

Jul 2024 - Present

- Own the Pre-Event Simulation process during race weekends within a rotating three-person team, delivering detailed pre-event reports to race engineers and senior management. Coordinate cross-departmental efforts to resolve bottlenecks and ensure timely delivery under race-weekend deadlines.
- Build data-driven setup recommendations by analysing ~2TB of simulation datasets for each race weekend, explicitly optimising risk reward trade-offs. Delivered quantified risk assessments with clear mitigation plans when assumptions diverged from reality.
- Reshaped the Pre-Event workflow and outputs by automating major components (Python, MATLAB), consolidating fragmented analyses and reducing reporting lead time from three weeks to one.
- Led a cross-department data integration project merging previously siloed car, simulation, and track metrics. Shipped a working MVP in three weeks, now used for model benchmarking, trend detection, and rapid miscorrelation diagnosis.
- Improved accuracy of car-balance targets by 70% (MSE reduction) by designing a new balance-target prediction method to replace a previously manual approach. This uncovered hidden miscorrelation patterns and enabled far more reliable pre-event decision-making.
- Designed and shipped visual analytics tools for spring selection, reducing setup complexity and enabling faster, clearer trade-off decisions during race weekends.

Agryq

London, England

Technical Co-founder

Jul 2025 – Present

- Built and shipped the full end-to-end MVP as technical co-founder for a personalised grant-matching platform for Spanish farmers, owning the entire stack from frontend to backend and AWS deployment. Implemented ethical web scraping, rule-based logic, ML classifiers, and constrained LLM pipelines with deterministic tooling and validation to ensure consistent outputs.
- Achieved strong early traction, onboarding several hundred farmers into the pilot programme shortly after launch.

Quasiscience

London, England

Core Developer

Nov 2022 – Mar 2024

- Created an application to analyse price trends of second-hand luxury items across 30 websites, collecting data on over 100 brands for the MVP, and generated key performance indicators (KPIs), including market volatility, price forecasts, and brand market share analysis.
- Initiated discussions with leading fashion brands to offer the tool as a B2B solution, as well as with Bloomberg to explore the potential for selling the data streams directly.

Mercedes-AMG Petronas Formula One Team

Brackley, England

Vehicle Dynamics Work Placement

Jul 2021 - Sep 2022

- In charge of finding the theoretical optimum aerodynamic setup prior to each race weekend.
- Analysed large data sets and reported key information to the strategists under a high level of pressure so the team can make informed decisions during the races.
- Enhanced the modelling capabilities of simulation tools and automate key cross-departmental tasks.
- Earned high praise and an extension of the internship to further maximize contributions to the team, a result of excellent problem-solving under pressure and consistently delivering high-quality work.

Education

University of Cambridge – Clare College

Cambridge, England

MEng in Manufacturing and Management Engineering (**First Class with Distinction**)

2019-2024

Relevant Modules: Mechanics and Vibrations, Mathematics and Computing, Advanced Operations Management, Leadership and management

Relevant Coursework

- 3rd Year Major Project Lead:** Directed both the engineering and financial aspects of developing a robotic pipe sealing device, taking it from concept to market.
- Industrial Placement Project Lead:** Directed development of an AI-based fault detection system that improved client product yields by 30%, leading to an additional £100,000 in revenue. Utilized Python to build a vision system that identified defect patterns.

Skills

Technical: Python, SQL, MATLAB, AWS, Azure DevOps, Data Pipelines, ML Modelling, Simulation and Optimisation, Kubernetes

Strengths: End-to-end ownership, Rapid prototyping, High-pressure execution, Cross-functional leadership, Risk analysis